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Case No. EC-2002-01038
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From: Sarah Novosel <sarah.novosel@enron.com>
Sent: Wed, 27 Dec 2000 03:44:00 -0800 (PST)
To: Steven J Kean <steven.kean@enron.com>, Richard Shapiro <richard.shapiro@enron.com>, Linda Robertson <linda.robertson@enron.com>, James D Steffes <james.steffes@enron.com>, Joe Hartsoe <joe.hartsoe@enron.com>, Sarah Novosel <sarah.novosel@enron.com>, Mary Hain <mary.hain@enron.com>, Tim Belden <tim.belden@enron.com>, Jeff Dasovich <jeff.dasovich@enron.com>, Mona L Petrochko <mona.petrochko@enron.com>, Sandra McCubbin <sandra.mccubbin@enron.com>, Karen Denne <karen.denne@enron.com> ...
Subject: Today's Van Vactor column

----- Forwarded by Sarah Novosel/Corp/Enron on 12/27/2000 11:42 AM -----

"Tabors, Richard" <tabors@tca-us.com>
12/27/2000 10:39 AM

To: dwatkiss@bracepatt.com, snovose@enron.com
cc:
Subject: Today's Van Vactor column

Dan and Sarah

I don't know if you see Sam VanVactor's materials or not. If not, here is a shot from the 26th

Richard

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EDITORIAL
S.A. VAN VACTOR
Energy Crisis Myths
Mythology has always been an important part of the human experience and never more so than when hard choices must be made. Thus it should come as no surprise that our current energy crisis is accompanied by a seemingly endless array of apologies, exaggerations, and myths.
Far and away the most common myth concerns the persistent view of many that suppliers are holding back generation in order to jack up prices. Similar charges were levied during the oil crises of the 1970s. In those

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days, critics complained that tankers could be seen just off the coast, waiting until prices rose. Holding back on electricity sales makes even less sense because once a moment of demand has passed, it is gone forever. No rational supplier will turn down the price offers persisting in the present marketplace day after day. Indeed, most suppliers are laboring to squeeze out every electron possible from aging and unreliable generators.

An equally common error arises when the crisis is blamed on "deregulation" and free markets. Although California was the first state to adopt wide-scale consumer choice, its market has been heavily regulated and constrained. For example, after nearly three years of consumer choice in the retail market only 12% of the total volume is served by entities other than traditional utilities. The volume of direct access has actually declined in recent months as buyers have scrambled to reconnect to regulated utilities that offer frozen rates. More importantly, the development of reliable forward markets and planning was stillborn because the California Public Utility Commission either prohibited or limited such trading at the same time that it granted automatic "prudency" to spot purchases. Deregulation may have changed the nature of the crisis, from a physical to a financial problem, but the demand and supply fundamentals at the heart of the current energy crises were inherited from an earlier era of regulatory stumbles.

Officials in neighboring states are eager to blame California for the crisis, but nothing could be further from the truth. The Western power market is fully interconnected by design and if one part is short, all regions are short. In fact, every operating area recognized that demand would occasionally exceed supply; planners simply relied on the proposition that it would not happen simultaneously in all regions of the West. However, this is exactly what has happened.

The current crisis was precipitated by a scarcity of natural gas, unexpected demand growth in California and the Southwest, the second hottest summer on record in the Southwest, and low water runs in Northern California and the Pacific Northwest.

If these myths are at odds with reality, how should the energy crisis be viewed? Famine and pestilence are normally reserved for biblical parables, but they do play

a role in this story. The Western power market is tied to the vast generation potential of the region's hydroelectric system. The system's potential is not, however, always available. Unlike hydro systems in Quebec and Brazil, the Columbia River system has only three months worth of water storage. (Quebec is said to have ten years of water inventory). The difference between a drought and a year flush with water can be as high as 6,000 average megawatts per hour - roughly the capacity of six of the West's nuclear generating stations. Thus, the availability of electricity swings from abundance to scarcity in an unpredictable pattern based on the state of the region's rivers. In the key months of June and July this year, hydroelectric generation from the Northwest Power Pool averaged 4,635 megawatts per hour below that of the favorable year of 1997.

System planners in the 1970s understood the nature of the hydroelectric system very well and planned to add thermal generation facilities to ensure adequate supply even under "critical-water" conditions. These conditions represented the worst-case scenario for hydroelectric generation. Such planning, however, renders most prospective thermal plants uneconomic, because they would lay idle except under the driest of conditions. With the collapse of the WPPSS plants, the Pacific Northwest quietly shifted its planning criteria from critical water to average water. In times of heavy water flow the region is a net exporter, but during a drought, it must import electricity.

Over the last decade, utility planning has been replaced by a market, and not just in California. Almost all new generators are independent power producers. In a market, new supply will only be built if there is the incentive to do so. That incentive takes the form of profits, and profits arise through high prices. What no one has thought through is the implications of the variability of hydro-electric power on the incentive to construct thermal generation. If new thermal generation is constructed, but is only needed one out of every ten years, then prices in years of drought must be extremely high in order to make building such thermal plants attractive to investors. In short, this is what the western power market is experiencing this year, and will continue to experience until water flows improve and new supplies catch up with the region's overall economic growth.

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The first step in solving a problem is to understand its cause. While it is true that California and other Western states have not invested sufficiently in generation capacity and natural gas delivery systems, additional

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